

USING THE BAYESIAN INFORMATION CRITERION TO OBTAIN TWO-STAGE DESIGNS ROBUST TO MODEL UNCERTAINTY

Arvind Ruggoo

Faculty of Agriculture, University of Mauritius, Reduit, Mauritius
e-mail: *aruggoo@uom.ac.mu*

and

Martina Vandebroek

Faculty of Economics and Applied Economics, University Centre for
Statistics, Katholieke Universiteit Leuven, Naamsestraat 69, B–3000 Leuven,
Belgium

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Summary: In this paper, we use the Bayesian Information Criterion (BIC) to develop a Bayesian two-stage design robust to model uncertainty. The BIC is particularly appealing in this situation as it avoids the necessity of specification of priors on the model parameters and can readily be computed from the output of standard statistical software packages.

1. Introduction

D-optimality and alphabetic optimality criteria in general have been criticized for being too dependent on the assumed model and for making no provision for model checking. Research in the recent years has concentrated on developing criteria that retain the flexibility of the D-optimal approach but also reduce dependence on model misspecification. See, for example, Steinberg and Hunter (1984), Wiens (1990), DuMouchel and Jones (1994), Chang and Notz (1996), Borkowski and Valeroso (2001) and the recent article by Heredia-

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Langner et al. (2004) for some extensive references in this area.

Another strategy to develop model-robust designs found in the literature is the use of sequential procedures. Neff (1996) developed Bayesian two-stage designs under model uncertainty for linear models. Lin, Myers and Ye (2000) developed Bayesian two-stage D-D optimal designs for models in experiments involving mixtures. Montepiedra and Yeh (1998, 2002) and Biswas and Chaudhuri (2002), amongst others, have also investigated sequential strategies for the linear model but for approximate designs. Ruggoo and Vandebroek (2004) (henceforth referred to as RUVA) have recently reviewed and extended the Bayesian two-stage D-D optimal approach of Neff (1996). See also Ruggoo and Vandebroek (2006) who have developed model-sensitive sequential optimal experimental designs.

In the two-stage procedure of RUVA, the first stage design is obtained using the Bayesian D-optimality criterion of DuMouchel and Jones (1994). The second stage D-optimal design is then generated from the first stage by incorporating improved model knowledge from the first stage experiment. However, the second stage design depends on the specification of prior model probabilities which are often hard to obtain.

The objective of this paper is to develop a two-stage procedure that is independent on the specification of prior distributions on the model parameters. We shall for that purpose use the Bayesian Information Criterion (BIC) which avoids the necessity of specification of a prior distribution on the model parameters and is useful for many practical purposes.

The paper will be organized as follows: in section 2, we set the notations to be used in the paper. We introduce the idea of Bayesian model averaging in section 3 and consequently illustrate that BIC can be used as an approximation

of the integrated likelihood. In section 4, we develop a BIC based Bayesian two-stage procedure that is robust to model uncertainty. We evaluate the new procedure in section 5 and compare it to alternatives available in the literature. Finally section 6 contains a brief conclusion.

2. The two-stage design procedure

Let us assume that the linear model that will be fitted by the experimenter is of the form

$$y = \mathbf{x}'_{pri} \boldsymbol{\beta}_{pri} + \varepsilon \quad (1)$$

with $\mathbf{x}_{pri}$ being a p -dimensional vector of powers and products of the factors and $\boldsymbol{\beta}_{pri}$ the p -dimensional vector of unknown parameters attached to the primary terms. Suppose that the expected response was misspecified so that the true model is actually of the form

$$y = \mathbf{x}' \boldsymbol{\beta} + \varepsilon = \mathbf{x}'_{pri} \boldsymbol{\beta}_{pri} + \mathbf{x}'_{pot} \boldsymbol{\beta}_{pot} + \varepsilon, \quad (2)$$

where $\mathbf{x}_{pot}$ is the q -dimensional vector containing powers and products of the factors not included in the fitted model and $\boldsymbol{\beta}_{pot}$ is the q -dimensional vector associated with the potential terms. Let $\mathbf{X}_{pri}$ be the $n \times p$ model matrix for the primary terms and $\mathbf{X}_{pot}$ the $n \times q$ model matrix for the potential terms. We further assume that $\mathbf{y}_i | \boldsymbol{\beta} \sim N(\mathbf{X}_i \boldsymbol{\beta}, \sigma^2 \mathbf{I})$ for each stage $i (i = 1, 2)$ with n_1 and n_2 observations in the first and second stage respectively so that the total number of observations in the entire experiment $n = n_1 + n_2$. $\mathbf{X}$ is the extended design matrix of dimension $n \times (p + q)$ for the combined stages, so that $\mathbf{X}' = [\mathbf{X}'_1 \mathbf{X}'_2]$. $\mathbf{X}_{pri(i)}$ and $\mathbf{X}_{pot(i)}$ correspond to the primary and potential terms respectively for each stage $i (i = 1, 2)$.

Before observing the first stage data, the experimenter has specified a set of $(p + q)$ regressors defining the full model. The true relationship between

the response and the input variables is believed to contain all primary terms and a subset q_i ($0 \leq q_i \leq q$) of the potential terms. Consequently the total number of possible models is $m = 2^q$. Let us consider the subset models $M_0, M_1, \dots, M_{m-1}$, with each model M_k defined by its corresponding parameters β_k , where M_0 corresponds to the primary terms model.

As mentioned earlier, the first stage design of RUVA is obtained as described in DuMouchel and Jones (1994). The second stage design is then obtained using posterior model probabilities computed from first stage data as weights in the design criterion (see Box and Meyer (1993) and RUVA for details). The two-stage design procedure of RUVA, is built within the realm of the Bayesian paradigm and consequently requires prior densities for the model parameters and also prior model probabilities. As is often the case, the problem of determining a prior distribution from available information is the most delicate matter in Bayesian methodology. Consequently, we shall develop a new two-stage procedure that is independent of the specification of priors. We shall for that purpose approximate the integrated likelihood, using the Bayesian Information Criterion (BIC).

3. Bayesian model averaging

The appealing characteristic of the Bayesian D-D optimal approach of RUVA is the incorporation of model uncertainty by averaging the criterion over all possible models in the second stage. In doing so, the procedure allows incorporation of several competing models and does not depend on specification of a single model. This approach can be thought of as a particular case of the Bayesian Model Averaging (BMA) procedure reviewed by Hoeting et al. (1999) and more recently by Clyde and George (2004). BMA provides a mechanism to account for model uncertainty by estimating some quantity

under each model and then averaging the estimates according to how likely the model is (Wasserman, 1997). Madigan and Raftery (1994) note that averaging over all models in this fashion provides better average predictive ability than using any single model.

Let us consider the subset models $M_0, M_1, \dots, M_{m-1}$ described earlier, with each model M_k defined by its corresponding parameters β_k . Each model consists of a set of probability densities for the random variable $\mathbf{y}$. Once we obtain first stage data $\mathbf{y}_1$, the posterior probability for model M_j can be easily evaluated from Bayes' theorem as follows:

$$p(M_j | \mathbf{y}_1) \propto p(M_j)p(\mathbf{y}_1 | M_j), \quad (3)$$

where $p(M_j)$ is the prior probability of model M_j and $p(\mathbf{y}_1 | M_j)$ is the integrated likelihood of $\mathbf{y}_1$ given model M_j .

From classical probability theory, $p(\mathbf{y}_1 | M_j)$ can be obtained by integrating over β_j , i.e.,

$$p(\mathbf{y}_1 | M_j) = \int p(\mathbf{y}_1 | \beta_j, M_j)p(\beta_j | M_j)d\beta_j,$$

where $p(\mathbf{y}_1 | \beta_j, M_j)$ is the likelihood function for model M_j . The quantity $p(\mathbf{y}_1 | M_j)$ is usually called the integrated likelihood for model M_j . Evaluating the integrated likelihood involves specifying priors for β_j and usually complex integration. The formulation of the prior and integration problems can be solved in the following manner for regular statistical problems: Let $\hat{\ell} = \ell_j(\hat{\beta}_j)$ denote the maximized log-likelihood under model M_j and d_j be the dimension of β_j . Then

$$\mathcal{S}_j = \hat{\ell}_j - \frac{d_j}{2} \log(n_1), \quad (4)$$

is a fairly accurate approximation of $\log p(\mathbf{y}_1 | M_j)$ for a specific choice of prior called the “unit-information prior” on the parameter space, that says that

the amount of information in the prior equals the amount of information in one observation (see Kass and Wasserman, 1995, for more details). Raftery (1996) gives further evidence for the accuracy of this approximation. We observe that expression (4) is the familiar Schwarz criterion (Schwarz, 1978) and minus twice the Schwarz criterion is often referred to as the BIC (Kass and Raftery, 1995), i.e.,

$$BIC_j = -2\mathcal{S}_j \approx -2 \log p(\mathbf{y}_1 | M_j). \quad (5)$$

From (5), we have an approximate but easy way of obtaining the integrated likelihood which does not depend on prior specification of the model parameters:

$$p(\mathbf{y}_1 | M_j) \approx e^{-0.5BIC_j}. \quad (6)$$

For the linear regression with normal errors, Raftery (1995) shows that the most convenient form of BIC is

$$BIC_j = n_1 \log(1 - R_j^2) + k_j \log(n_1), \quad (7)$$

where R_j^2 is the usual R^2 (coefficient of determination) value for model M_j and k_j is the number of regressors (not including the intercept) in the model. Using our results from (6) and (7), the posterior probability of each model is easily found from (3) to be

$$p(M_j | \mathbf{y}_1) = C_1 p(M_j) e^{-0.5BIC_j}, \quad (8)$$

where C_1 is a normalizing constant that forces all posterior probabilities to sum to unity. As explained by Kass and Wasserman (1995), the BIC uses an implicit unit information prior, i.e., a multivariate normal prior with mean at the maximum likelihood estimate and the amount of information in the prior equal to the average amount of information in one observation. Since the prior is based on only one observation, it is vague yet proper. Further it involves readily available regression statistics for all candidate models which

can be obtained from the output of standard statistical software packages. The appealing property of BIC, which avoids prior specification on model parameters, has also been investigated by Pauler (1998) who motivates the BIC and proposes two useful modifications of the criterion applicable to other types of problems. Volinsky and Raftery (2000) have investigated BIC for variable selection in models for censored survival data.

Wasserman (1997) also shows that if model M_j denotes the model containing the true density, i.e. the model that generates the data, then for $i \neq j$ and under weak conditions,

$$\frac{p(M_i | \mathbf{y}_1)}{p(M_j | \mathbf{y}_1)} \longrightarrow 0$$

in probability. This means that the posterior probability of the true model goes to one and the posterior probabilities of the other models go to zero. Further Wasserman (1997) indicates that the BIC has the same asymptotic behaviour as it selects the true model asymptotically.

4. Development of the two-stage procedure using the BIC

In the first stage, the ‘true’ model is unknown and the investigator would like to refine the primary model entertained. So a criterion that would include the ability to test for lack of fit would be desirable. The first stage design can then be obtained using a sort of weighted combination of the D-optimum design for β_{pri} and the D_s -optimum design for β_{pot} suggested by Atkinson and Donev (1992):

$$D_1 = \frac{1}{p} \log \left| \left(\mathbf{X}'_{pri(1)} \mathbf{X}_{pri(1)} \right)^{-1} \right| + \frac{1}{q} \log |\mathbf{L}_1|^{-1}, \quad (9)$$

where

$$\mathbf{L}_1 = \mathbf{X}'_{pot(1)} \mathbf{X}_{pot(1)} - \mathbf{X}'_{pot(1)} \mathbf{X}_{pri(1)} (\mathbf{X}'_{pri(1)} \mathbf{X}_{pri(1)})^{-1} \mathbf{X}'_{pri(1)} \mathbf{X}_{pot(1)}$$

is usually referred to as the dispersion matrix and is ubiquitous in the literature on model-sensitive designs. The dual-purpose criterion ensures efficient estimation of parameters of the assumed primary model and detection of departures from that model. It is also interesting to recognize that the composite design criterion (9) of Atkinson and Donev (1992) does not depend on any form of prior assumption on the model parameters, β . Goos et al. (2005) have recently established the similarity between the criterion of DuMouchel and Jones (1994) and that of Atkinson and Donev (1992).

Therefore we can obtain our first stage design $\mathbf{X}_1 = [\mathbf{X}_{pri(1)} \mathbf{X}_{pot(1)}]$ by minimizing the criterion in (9). In practice while using (9), it is often observed that L_1 might be singular. The problem is avoided in the algorithmic construction of designs by the addition of a small multiple of the identity matrix. That is, we let

$$\mathbf{L}_1 = \mathbf{L}_1 + \nu \mathbf{I}_q, \quad (10)$$

where ν is a small number typically between 10^{-4} and 10^{-6} . Such type of matrix regularization is common in the construction of exact D-optimal designs (see for e.g., Atkinson and Donev, 1992, chapters 10 & 15). In the second stage, we can again use the idea of Atkinson and Donev (1992), so that the second stage design for model $M_k (k = 1, 2, \dots, m-1)$, is the set of design points which minimizes

$$D_2^{(k)} = \frac{1}{p} \log \left| \mathbf{X}_{pri(1)}^{(k)'} \mathbf{X}_{pri(1)}^{(k)} + \mathbf{X}_{pri(2)}^{(k)'} \mathbf{X}_{pri(2)}^{(k)} \right|^{-1} + \frac{1}{q} \log \left| \mathbf{L}_2^{(k)} \right|^{-1} \quad (11)$$

where

$$\mathbf{L}_2^{(k)} = \left[\left(\mathbf{X}_{pot(1)}^{(k)'} \mathbf{X}_{pot(1)}^{(k)} + \mathbf{X}_{pot(2)}^{(k)'} \mathbf{X}_{pot(2)}^{(k)} \right) - \left(\mathbf{X}_{pot(1)}^{(k)'} \mathbf{X}_{pri(1)}^{(k)} + \mathbf{X}_{pot(2)}^{(k)'} \mathbf{X}_{pri(2)}^{(k)} \right) \times \right. \\ \left. \left(\mathbf{X}_{pri(1)}^{(k)'} \mathbf{X}_{pri(1)}^{(k)} + \mathbf{X}_{pri(2)}^{(k)'} \mathbf{X}_{pri(2)}^{(k)} \right)^{-1} \left(\mathbf{X}_{pri(1)}^{(k)'} \mathbf{X}_{pot(1)}^{(k)} + \mathbf{X}_{pri(2)}^{(k)'} \mathbf{X}_{pot(2)}^{(k)} \right) \right],$$

is the dispersion matrix for the combined design for model $\mathbf{M}_k$.

$\mathbf{X}_{pri(1)}^{(k)}, \mathbf{X}_{pri(2)}^{(k)}, \mathbf{X}_{pot(1)}^{(k)}, \mathbf{X}_{pot(2)}^{(k)}$ are the matrices corresponding to $\mathbf{X}_{pri(1)}, \mathbf{X}_{pri(2)}, \mathbf{X}_{pot(1)}$ and $\mathbf{X}_{pot(2)}$ expanded to model space $\mathbf{M}_k$.

We can again avoid singularity problems for $\mathbf{L}_2^{(k)}$ using the idea in (10). Since $p(M_k | \mathbf{y}_1)$ in (8), reflect model importance, they can be incorporated as measures of fit to average the design criterion. Consequently, the second stage design is then obtained by choosing the second stage design points $\mathbf{X}_2 = [\mathbf{X}_{pri(2)} \mathbf{X}_{pot(2)}]$ so as to minimize

$$\sum_{M_k} p(M_k | \mathbf{y}_1) D_2^{(k)}.$$

The modifications in the two-stage procedure implicitly avoid specification of a prior distribution on the model parameters in the optimality criterion and provide, as we shall see later, an attractive alternative to the approach of RUVA.

5. Evaluation of the procedure

To enable comparison as to how our BIC based two-stage procedure compares to some of the approaches in the literature, we shall consider the three cases below. The competitors to the BIC based two-stage designs are the Bayesian D-D optimal designs of RUVA and the one-stage traditional D-optimal design for the full model and the one-stage Bayesian D-optimal design for all $(p + q)$ terms under consideration. We shall use equal sample size partitioning in the two stages, with $n_1 = (p + q + 2) = n_2$, as suggested by RUVA and Neff and Myers (1998).

Case I:

We consider a model with $p = 5$ primary terms, $\{1, x_1, x_2, x_3, x_1x_2\}$ and $q = 4$ potential terms, $\{x_1^2, x_1x_3, x_2^2, x_3^2\}$. Here $n_1 = n_2 = 11$. First stage data is then simulated from

$$y = 42.0 + 11.2 x_1 + 14.5 x_2 + 10.6 x_3 + 12.5 x_1x_2 + 8.9 x_1^2 - 9.9 x_1x_3 + \varepsilon.$$

Case II:

In this case, we examine data simulated from

$$y = 40.0 + 11.5 x_1 + 12.8 x_2 + 10.5 x_3 + 14.6 x_1^2 + 9.8 x_1 x_2 - 7.4 x_1 x_3 - 8.7 x_2^2 + \varepsilon$$

and with the full model comprising of five primary terms, namely $\{1, x_1, x_2, x_3, x_1^2\}$ and five potential terms, $\{x_1 x_2, x_1 x_3, x_2 x_3, x_2^2, x_3^2\}$. Also $n_1 = n_2 = 12$.

Case III:

In this case we consider $p = 7$ primary terms, $\{1, x_1, x_2, x_3, x_1 x_2, x_1^2, x_2^2\}$ and $q = 6$ potential terms, $\{x_4, x_1 x_3, x_2 x_3, x_1 x_4, x_3^2, x_4^2\}$. First stage data is simulated from the true model with the seven primary terms and three of the six potential terms:

$$\begin{aligned} y = & 30.0 + 12.8 x_1 - 9.2 x_2 + 12.0 x_3 + 11.9 x_1 x_2 - 8.2 x_1^2 + 10.3 x_2^2 \\ & - 6.0 x_4 + 4.4 x_2 x_3 + 5.2 x_3^2 + \varepsilon. \end{aligned}$$

There are resources for 30 runs and an equal partitioning is used in the two stages.

The performance of each design is measured by its design efficiency relative to regressors in the true model. The measure of efficiency is the usual D-optimality criterion, namely

$$D^* = \left| n (\mathbf{X}^{*'} \mathbf{X}^*)^{-1} \right|$$

where $\mathbf{X}^*$ contains the $n = (n_1 + n_2)$ design points and is expanded to contain regressors present in the true model. Since the two-stage procedure is dependent on first stage data, the performance of the two-stage procedures is measured by the average D^* over 200 different simulations, i.e.,

$$\overline{D}^* = \frac{\sum_{i=1}^{200} D_i^*}{200}.$$

We assume that $\varepsilon \sim N(0, 1)$ in all the simulations conducted. The standard deviations of D^* are also recorded over the numerous simulations. The one-stage traditional non-Bayesian D-optimal design and one-stage Bayesian D-optimal design are not data dependent and can thus be evaluated by a single D^* , namely $D^* = \left| n (\mathbf{X}^*{}' \mathbf{X}^*)^{-1} \right|$ for regressors present in the true models.

Table 1. Comparison of the two-stage and the single stage design procedures

Case I. $y = 42.0 + 11.2 x_1 + 14.5 x_2 + 10.6 x_3 + 12.5 x_1 x_2 + 8.9 x_1^2 - 9.9 x_1 x_3 + \varepsilon$

Two-stage approach	RUVA	0.0318 (0.00075)
	BIC	0.0363 (0.01102)
One-stage approach	D-optimal for full model	0.0602
	DuMouchel & Jones (1994)	0.0550

Table 2. Comparison of the two-stage and the single stage design procedures

Case II. $y = 40.0 + 11.5 x_1 + 12.8 x_2 + 10.5 x_3 + 14.6 x_1^2 + 9.8 x_1 x_2 - 7.4 x_1 x_3 - 8.7 x_2^2 + \varepsilon$

Two-stage approach	RUVA	0.0344 (0.00193)
	BIC	0.0381 (0.00511)
One-stage approach	D-optimal for full model	0.0515
	DuMouchel & Jones (1994)	0.0467

Table 3. Comparison of the two-stage and the single stage design procedures

Case III. $y = 30.0 + 12.8 x_1 - 9.2 x_2 + 12.0 x_3 + 11.9 x_1 x_2 - 8.2 x_1^2 + 10.3 x_2^2 - 6.0 x_4 + 4.4 x_2 x_3 + 5.2 x_3^2 + \varepsilon$.

Two-stage approach	RUVA	0.0286 (0.00140)
	BIC	0.0303 (0.00114)
One-stage approach	D-optimal for full model	0.0334
	DuMouchel & Jones (1994)	0.0324

The results of the evaluations for all the cases are shown in Tables 1 to 3. The two-stage procedures use the simple but powerful idea of incorporating previous observations to guide the choice of new design points. The benefit of using previous observations in the two-stage approaches is evident from Tables 1 to 3, with better precision of parameters of the true model than the one-stage competitors in the three cases considered. Using the BIC as an approximation to the marginal likelihood is reassuring as it gives very good and comparable results to the ones obtained by RUVA. The injection of additional prior information and consequently more precise posterior probabilities in the procedure by RUVA may account for the smaller variances in the different cases and also slightly better average D-values than using the BIC based procedure.

6. Conclusion

In this paper, we have used the BIC in the development of a two-stage procedure that makes it possible to efficiently design experiments when initial knowledge of the regressors of the model is poor. The idea of using the BIC as an approximation to the integrated likelihood is borrowed from tools commonly used in the flourishing Bayesian model averaging literature. We are currently unaware of any design procedure that explicitly uses the BIC to attack model uncertainty in experimental design problems in this way. The fact that computation of the BIC does not require introduction of prior distributions on the model parameters and relies on the unit-information prior which is the average amount of information in one observation, makes it intuitively appealing.

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